

Week 14 Pre-Lab Quiz — Bootstrap Confidence Intervals

CE 310 | 10 questions | 10 points | Due before lab session

Q1. What distinguishes bootstrap resampling from Monte Carlo simulation (Week 13)?

- A. Bootstrap samples from the data itself (with replacement); Monte Carlo samples from assumed parametric distributions
- B. Bootstrap requires more samples than Monte Carlo to converge
- C. Monte Carlo resamples the data; bootstrap assumes a normal distribution
- D. They are the same method with different names

Q2. The sample mean Outdoor_Temp_F is $\bar{x} = 68.745^\circ\text{F}$, with an analytical standard error $\text{SE} = 0.1808^\circ\text{F}$. Without using Python, compute the 95% analytical confidence interval using $\bar{x} \pm 1.96 \times \text{SE}$.

- A. (68.391, 69.099) — correct: $68.745 \pm (1.96 \times 0.1808) = 68.745 \pm 0.3544$
- B. (68.564, 68.926) — uses $\pm 1.0 \times \text{SE}$ instead of $\pm 1.96 \times \text{SE}$
- C. (67.745, 69.745) — misreads SE as a flat $\pm 1.0^\circ\text{F}$ margin
- D. (68.745, 69.099) — only adds the margin on the upper side

Q3. You compute 1,000 bootstrap means of outdoor temperature and take the 2.5th and 97.5th percentiles. This is called:

- A. The BCa (bias-corrected and accelerated) bootstrap confidence interval
- B. The percentile bootstrap 95% confidence interval
- C. A one-sided 97.5% confidence interval
- D. A t-interval corrected for skewness

Q4. Your bootstrap SE for the mean outdoor temperature is 0.179°F , and the analytical SE ($\sigma/\sqrt{n}$) is 0.181°F . What does this suggest?

- A. The bootstrap is incorrect — it should always match the analytical SE exactly
- B. The data are not normally distributed, so bootstrap should be used
- C. Both methods give nearly identical results here; bootstrap's advantage is for complex statistics, not the mean
- D. The analytical SE is wrong because it assumes independence

Q5. You bootstrap 1,000 regression slopes for $\text{Cooling} \sim \text{Outdoor_Temp_F}$. The bootstrap 95% CI is [1.132, 1.181], while the analytical OLS 95% CI is [1.139, 1.178]. The bootstrap CI is slightly wider. What is the most likely explanation?

- A. The bootstrap has sampling error; run more replications
 - B. The bootstrap reflects temporal autocorrelation in hourly data that OLS standard errors ignore
 - C. The OLS CI is always too narrow because OLS assumes normality
 - D. The bootstrap slope was computed incorrectly
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Q6. A permutation test for the BH effect on cooling shuffles the BH labels 1,000 times and measures the mean difference each time. None of the 1,000 permutations produces a difference as large as the observed 28.3 kWh/hr. What is the permutation p-value?

- A. 1.0 (all permutations exceed the observed)
 - B. 0.028 (28 permutations out of 1,000)
 - C. 0.0 (zero permutations match or exceed the observed)
 - D. 0.5 (the null hypothesis predicts a 50% chance)
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Q7. You estimate $P(\text{Cooling} > 30 \text{ kWh/hr}) = 0.265$ and bootstrap a 95% CI of [0.256, 0.275]. A colleague computes a Wilson proportion CI of [0.256, 0.274] for the same proportion. What is the correct interpretation?

- A. Both give essentially the same interval; bootstrap is unnecessary here
 - B. The Wilson CI is invalid for continuous data
 - C. The bootstrap CI is always preferred because it makes no assumptions
 - D. You need at least 10,000 bootstrap samples before the two methods agree
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Q8. For which of these statistics is bootstrap CI clearly preferable to an analytical approach?

- A. Sample mean with $n = 10,000$ normally distributed observations
 - B. Median of a heavily skewed distribution with $n = 40$
 - C. Proportion from a binomial distribution with $n = 1,000$
 - D. Difference between two means from independent normal populations
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Q9. A pairs bootstrap for regression (resample rows together) is preferred over residual bootstrap (fix predictors, resample residuals) when:

- A. The residuals are normally distributed
 - B. The sample size is large
 - C. The predictor values are not fixed by design — you want to capture uncertainty in both X and Y
 - D. You want a narrower confidence interval
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Q10. You increase the number of bootstrap resamples from 1,000 to 10,000. What changes?

- A. The bootstrap CI becomes narrower (more resamples = more precision in the statistic)
 - B. The bootstrap CI endpoints become more stable (less Monte Carlo error in the percentiles), but the width reflects data variability and does not shrink
 - C. The bootstrap mean changes significantly
 - D. The computation time decreases because each resample is smaller
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